PID/DATA MONITOR TABLE [EPS CONTROL MODULE]

id060230813400

Note
- If the PID “STR_ANG” is inspected, perform the following procedure before inspecting.
  1. Switch the ignition to OFF, and maintain this condition for **approx. 3 s**.
  2. Set the vehicle wheels in the straight-ahead position.
  3. Switch the ignition ON (engine off or on).

—: Not applicable

Monitor item M-MDS display	Unit/ Operation	Test condition	Specification (Reference)	Inspection item
ECU_IN_TMP	°C, °F	This PID indicates temperature of board in EPS control module.		• Replace the steering column. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
ENG_RPM	RPM	This PID indicates engine speed.		• Inspect the PCM. (See PCM INSPECTION [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See PCM INSPECTION [SKYACTIV-D 1.5].)
MT_CURRENT	A	When not steered	Near 0 A	• Replace the steering column. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
		Steered left	Changes to 0 A— negative	
		Steered right	Changes to 0 A— positive	
OH_CR_C	Yes/No	Overheat protection control (complete) is operating	Yes	—
		Overheat protection control (complete) is not operating	No	
OH_CR_M	Yes/No	Overheat protection control (middle) is operating	Yes	—
		Overheat protection control (middle) is not operating	No	
OH_HIST_C	Yes/No	Before overheat protection control (complete) has operated	Yes	—
		Before overheat protection control (complete) has not operated	No	
OH_HIST_M	Yes/No	Before overheat protection control (middle) has operated	Yes	—
		Before overheat protection control (middle) has not operated	No	
OH_IG_CNT_C	—	The number of times the ignition is switched ON during the period from when overheat protection control (complete) is finished to the time the next overheat protection control (complete) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (complete) is started. • During overheat protection control (complete), the number of times the ignition is switched ON is not counted, and 0 is displayed.		—

Monitor item M-MDS display	Unit/ Operation	Test condition	Specification (Reference)	Inspection item
OH_IG_CNT_M	—	The number of times the ignition is switched ON during the period from when overheat protection control (middle) is finished to the time the next overheat protection control (middle) is started • Displays a maximum number of 255 times in which the ignition is switched ON until the first overheat protection control (middle) is started. • During overheat protection control (middle), the number of times the ignition is switched ON is not counted, and 0 is displayed.		—
STR_AB_EST **Note** • The signal indicated in this item is a steering angle (estimated absolute angle).	° ° °	Steering wheel position when vehicle wheels straight-ahead	0 °	• Perform the DTC inspection for the PCM, DSC HU/CM, and EPS control module. If a DTC is displayed after performing the DTC inspection for the PCM, DSC HU/CM, and EPS control module, repair the malfunctioning part according to the applicable DTC troubleshooting. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC INSPECTION [DSC HU/CM].) (See DTC INSPECTION [EPS CONTROL MODULE].) • After performing the DTC inspection, perform the following procedures: — Switch the ignition off and maintain condition for approx. **3 s**. — Switch the ignition ON (engine off or on). — Set the vehicle wheels straight-ahead, then drive the vehicle straight at 20 km/h {12 mph} or more for 5 s or more. — Using the M-MDS, verify the STR_ANG_EST operation condition. • If an abnormal value is indicated after verifying the value of the STR_AB_EST ANG again, replace the steering column. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
		Steered left	Changes to 0 °—positive	
		Steered right	Changes to 0 °—negative	

Monitor item M-MDS display	Unit/ Operation	Test condition	Specification (Reference)	Inspection item
STR_ANG **Note** • The signal indicated in this item is a steering angle (relative angle).	°	Steering wheel position when ignition is switched ON (engine off or on)	0 °	• Perform the DTC inspection for the PCM, DSC HU/CM, and EPS control module. If a DTC is displayed after performing the DTC inspection for the PCM, DSC HU/CM, and EPS control module, repair the malfunctioning part according to the applicable DTC troubleshooting. (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-G 1.3, SKYACTIV-G 1.5].) (See ON-BOARD DIAGNOSTIC TEST [SKYACTIV-D 1.5].) (See DTC INSPECTION [DSC HU/CM].) (See DTC INSPECTION [EPS CONTROL MODULE].) • After performing the DTC inspection, perform the following procedures: — Switch the ignition off and maintain condition for approx. **3 s**. — Set the vehicle wheels straight-ahead. — Switch the ignition ON (engine off or on). — Using the M-MDS, verify the STR_ANG operation condition. • If an abnormal value is indicated after verifying the value of the STR_ANG again, replace the steering column. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
		Steered left	Changes to 0 °—negative	
		Steered right	Changes to 0 °—positive	
STR_ROT_SPD	°/s	Not steered	Near 0 °/s	• Replace the steering column. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
		Steered	Changes according to steering speed	
STR_TRQ_S_M	Nm	Not steered	Near 0 Nm	• Replace the steering column. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
		Steered left	Changes to 0 Nm—positive	
		Steered right	Changes to 0 Nm—negative	
STR_TRQ_S_S	Nm	Not steered	Near 0 Nm	• Replace the steering column. (See STEERING WHEEL AND COLUMN REMOVAL/ INSTALLATION.)
		Steered left	Changes to 0 Nm—positive	
		Steered right	Changes to 0 Nm—negative	

Monitor item M-MDS display	Unit/ Operation	Test condition	Specification (Reference)	Inspection item
VPWR	V	This PID indicates power supply voltage for EPS control module.		• Inspect the battery. (See BATTERY INSPECTION.) • Inspect power supply circuit (ignition switch, IG1 relay, fuse etc.) (See PUSH BUTTON START REMOVAL/INSTALLATION.) (See RELAY INSPECTION.)
VSPD	KPH, MPH	This PID indicates vehicle speed.		• Inspect the DSC HU/CM. (See DSC HU/CM INSPECTION.)